

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP7-General Insurance Reserving and Capital Modelling

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q.1)

- i) Possible reasons why the management has kept the reinsurance arrangement unchanged:
- The insurer might have been small and new entrant into the business. Proportional reinsurance helps spread the risk.
 - Administratively simple for the insurer compared to other types of reinsurance.
 - The commission offered by the reinsurer is attractive.
 - The insurer has probably been writing risks of similar size and variance.
 - Other types of reinsurance are probably unavailable or unaffordable.
 - The business is being ceded to a captive reinsurer; hence, the arrangement is purely strategic driven.
 - The reinsurer might be offering additional services around underwriting, claims handling, new product design etc.
 - Regulator might have stipulated to take quota share reinsurance for stipulated amount of time for new insurers.
 - Strategic driven, the reinsurance might have been sourced by composite broker who is ceding bulk of direct business at a cheaper cost of acquisition.
 - The insurer might not have evaluated the reinsurance from a cost vs benefit perspective so far.

(Max 4)

ii) Reasons why the current structure may not be suitable:

- Size of risk and volatility:
 - i. The company may have grown in size and may have plans to venture into riskier and more volatile pockets of home insurance business.
 - ii. A surplus or individual excess of loss reinsurance can be more beneficial in such situations.
 - iii. The company might have excess capital available at a cheaper cost compared to quota share reinsurance.
 - iv. Plans to target insuring high-value residential properties, which can increase volatility to results.
- Geographical location and accumulation:
 - i. Exposure to areas with higher chance of catastrophes, would benefit from having catastrophe reinsurance.
 - ii. The company might have plans to write more houses in a specific location or community, prone to accumulation.
- Cost of reinsurance:
Since the business is profitable, a non-proportional reinsurance might be more optimal from cost vs benefit perspective.
- Change in regulatory environment:
 - i. Regulations might have changed to make some types of reinsurance compulsory.
 - ii. There could be move towards risk-based capital, where non-proportional reinsurance might show better benefit.
- Financial objectives:
Non-proportional reinsurance may be useful if the expectation of shareholders is to smooth profits over time.

(Max 5)

iii) Assumption: Reinstatement premium is earned uniformly throughout the year.

a) Outward cashflows for the insurer under both structures are:

	Reinsurer A	Reinsurer B
Reinsurance Premium	$20\% \times 30 + 15\% \times 15 = 8.25$	$40\% \times 35 = 14$

Claim1 less reinsurance recovery	$42 - (30 + 2) = 10$	$42 - 34 = 8$
Reinstatement Premium	$250\% * 20\% * 30 * 0.75 = 11.25$	$40\% * 34 * 0.75 = 10.2$
Claim2 less reinsurance recovery	$20 - 10 = 10$	$20 - 12 = 8$
Total Outgo	39.5	40.2

Hence, Reinsurer A's structure is better for the insurer.

b) Outward cashflows for the insurer under both the structures are:

	Reinsurer A	Reinsurer B
Reinsurance Premium	$20\% * 30 + 15\% * 15 = 8.25$	$40\% * 35 = 14$
Claim less reinsurance recovery	$35 - 25 = 10$	$35 - 27 = 8$
Reinstatement Premium	$250\% * 20\% * 25 * 0.5 = 6.25$	$40\% * 27 * 0.5 = 5.4$
Total Outgo	24.5	27.4

Hence, Reinsurer A's structure is better in this case as well.

(Max 8)

[17 Marks]

Q. 2)

i) Multi-year motor insurance underwriting

Multi-year policies involve guaranteeing premium for additional coverage period beyond one year. They differ from single year motor insurance policy in several ways:

- Assumptions need to be made for claim experience during the years, and corresponding impact on no-claims-bonus for each year.
- No lapses as premium is paid for multiple years, leading to lower acquisition cost due to ability to spread across multiple years.
- Higher chance of selective lapses as policyholders with good experience could purchase new covers from market at potentially lower price.
- Possible anti-selection due to guaranteed premium for future years.
- Greater uncertainty of future claims inflation, assumptions to be included in original price.
- Inflation assumption for liability claims can be more uncertain given the longer run off period.
- Greater investment income as premium for multiple years is received in advance and can be suitably invested for longer tenure.
- More capital intensive due to higher insurance risks stemming from longer term assumptions.

(Max 3)

ii) Benefits of regulatory changes

To insurer ABC

- Increase in investor confidence...
- ... leading to access to cheaper capital.
- Can take advantage of unrealised gains from market valuation of assets and liabilities.
- Better transparency leading to easier comparison with peers....
- ... rewarded for maintaining underwriting discipline.
- ... rewarded over peers for maintaining better reserving practices.

To insurer ABC's policyholders

- Greater transparency of results, leading to increase in trust

- Improved policyholder security and consumer protection
- Higher disclosure requirements would allow greater transparency on customer satisfaction; for example: claims rejection ratio, claims settlement delay, renewal ratio etc.....
- ...enabling policyholders to compare and make informed decisions when buying insurance cover.
- Leading to greater responsibility from insurance companies to build the right products for the target market.

To the regulator of Country Y

- Greater transparency, improving reliability of the industry in the long term...
- ...resulting in greater capital inflows and new entrants, improving insurance penetration.
- Gradual profit recognition, greater underwriting responsibility.
- Exposure of potential loss-making areas and ensure action is taken, leading to improvement in overall solvency.
- Granular assumption setting can lead to improvement in actuarial modelling technique...reducing in cross subsidy/distribution risk.

(Max 6)

iii) Financial impact of new regime

- Market value of assets and liabilities
 - Market valuation of certain assets might be difficult/lead to significant swings
 - Inability to find representative portfolio for market valuation of liabilities leading to imperfect value generation, especially for unique products like multi-year insurance.
 - Market value stressed due to significant negative influence of branding due to parent or partner company.
 - Economic situations can lead to market downturn as well as increase in claims at the same time.
 - Increase in market risk compared to before.
- Difference in recognition of profit and loss
 - Shift in profitability of existing business under new regime, possibly challenging key business segments.
 - Shocks to deal with loss making businesses immediately rather than over time
 - Delayed profit recognition for businesses like long tailed liability where runway could be lengthy
 - Increased underwriting scrutiny leading to fall in business volumes especially for riskier products with longer duration...
 - ...leading to difficulty in covering fixed expenses
- Granular assumption setting and disclosures
 - Increased uncertainty and complexity due to detailed modelling
 - lack of proper data for detailed assumptions
 - Greater disclosure requirements leading to potential loss of competitive advantage in assumption setting for it's multi-year motor product.
 - Increase in regulatory interference and scrutiny.
 - Lesser risk-taking ability – focus more on business that has higher chances of profits or familiar risks leading to drop in innovation. As a result, the company might choose to switch to annual vanilla motor product, leading to loss of competitive edge.
 - Difficulties to meet regulatory deadlines, leading to fines
- Risks and uncertainty
 - Increase in overall risks due to uncertainty – insurance risk, Operational risks etc.
 - Increase in cost of reinsurance
- Improved comparability and transparency:
 - Other companies could be perceived to be more attractive; this could lead to drop in share price or increase in cost of capital

- Comparison could be with peers who might have different business mix altogether
- Transparency in results may lead to incorrect interpretation of long-term initiatives which have higher cost in short term
- Comparison of multi-year with single year products can lead to incorrect conclusions
- Other negative impacts:
 - Additional costs in training employees to the new regime, lack of talent in the market
 - Incorrect decisions by senior management due to incomplete/inadequate understanding
 - Overhaul of IT systems, purchase of new software, mismatch between two regimes during implementation
 - Negative impact on next year's business plan due to uncertainty

(Max 7)

[16 Marks]

Q. 3)

- i) Market risk is defined as the risk that, as a result of market movements, a firm may be exposed to fluctuations in the value of its assets or the income from its assets. This risk exists to the extent that any movement in assets is not matched by a corresponding movement in the liabilities. (1)

- ii) Sources of general market risk include movements in:

- interest rates
- exchange rates
- equity prices
- property prices.

(Max 2)

- iii) Factors to consider when modelling market risk include:

- Changed market values of investments
- variation in interest rates and the effect on
 - the market value of investments
 - the value of liabilities, if discounting is applied
- level of investment income
- counterparty / issuer defaults (this might be addressed under credit risk)
- severe economic or market downturn or upturn leading to adverse interest rate movements and/or equity market falls
- inadequate valuation of assets (this might be addressed under operational risk)
- currency movements
- the extent of any mismatch of assets and liabilities, including reinvestment risk.

(Max 4)

- iv) An increase in capital charge could be due to:

1. Counterparty credit risk:
Increased risk of reinsurance default if the new reinsurer has a lower credit rating.
There could also be increased risk of reinsurance failure in extreme loss scenarios for the new reinsurer if the reinsured risks are not well diversified.
2. Liquidity risk:
Increased liquidity risk due to delays in collecting reinsurance recoveries, especially since the relationship is new.
3. Net insurance risk:
The reinsurance might have some exclusions that lead to lower recoveries in some scenarios.

4. Operational risk:
Lower cover / missed layers compared to before due to mis-purchase
Data errors due to new system / bordereaux requirements
5. Group risk:
Reputational risk or group reinsurance risk due to change of reinsurer if the earlier reinsurer had ties with other entities of the company or the group.
6. Strategic risk:
The chosen reinsurance program may be inappropriate for the company's future business plans
7. Regulatory risk:
The insurer might be switching from a domestic to an international reinsurer. This might attract additional regulatory scrutiny.

(Max 6)

v) Considerations while modelling:

1. Counterparty credit risk:
 - a) Dependency between different reinsurers of the company defaulting
 - b) Higher loss when default occurs due to lower credit rating
 - c) Non-recoveries due to reinsurance disputes, especially if the new reinsurer has lesser experience or has historical disputes with other insurers
2. Liquidity risk:
 - a) Delays in reinsurer reconciling payments and Bordeaux information
 - b) Extent of delay in reinsurance settlement after a large loss
3. Net insurance risk:
 - a) Changes in modelling future reinsurance recoveries compared to past recoveries due to change of reinsurer
 - b) Changes in recoveries of existing claim disputes due to change of reinsurer
4. Operational risk:
 - a) Incorrect interpretation of coverage due to changes in reinsurance wording compared to before
 - b) Complexity in implementation of new reinsurance structure into the system
5. Group risk:
 - a) Gaps with existing group reinsurance arrangements
 - b) Existing relations of earlier reinsurer with other entities of the group
6. Strategic risk:
The new reinsurer, while quoted lower rates, could have quoted terms that are not appropriate to the company's business plan.
7. Regulatory risk:
Additional delays in regulatory filing. Penalties for potential violation of domestic regulations if any. Implications if regulator rejects reinsurance policy.

(Max 6)

[Max 19 Marks]

Q. 4)

- i) The incurred but not enough reported (IBNER) reserve, is a reserve reflecting expected changes in estimates for claims reported to the (re)insurer by the valuation date only (that is, excluding any "true" or "pure" IBNR claims).

By contrast, the pure incurred but not reported (IBNR) reserve provides for amounts in respect of claim events that have occurred before the valuation date but are still to be reported to the (re)insurer by that date.

In the case of a reinsurer, the pure IBNR reserve also needs to provide for claims that, although already known to the cedant, have not yet been reported to the reinsurer as being liable to involve the reinsurer.

(Max 1)

ii) Ultimate claim count for each accident year

Step1: Calculate the development factor for each duration

Development Period	1	2	3	4	5
Development Factor	2	1.8	1.25	1.05	1

For example, development factor for reported delay 3 is calculated as claims reported as at delay 3 divided by claims reported as at delay 2 i.e.

$$\frac{710+850+800+1600}{350+550+400+900} = \frac{3960}{2200} = 1.8$$

Step2: Calculate the cumulative development factor and Ultimate Claims Count for each accident year

Accident Year	Count of claims reported to date	Cumulative development factor	Ultimate Claims Count
2018	1000	1	1000
2019	1100	1	1100
2020	950	1.05	998
2021	1600	1.31	2100
2022	1000	2.36	2363
2023	250	4.73	1181

For example, cumulative development factor is calculated as cumulative product of each development factor, assuming the first accident year is fully developed.

For 2021, it is calculated as $1 \times 1 \times 1.05 \times 1.25 = 1.31$

Ultimate claim count is calculated as count of claims reported to date multiplied by the respective cumulative development factor.

For 2021, it is calculated as $1600 \times 1.31 = 2100$

(Max 5)

iii) Pure IBNR and IBNER estimation

Step1: Derive IBNR claims count as Ultimate claim count minus claims reported till date.

Accident Year	Ultimate Claim Count (a)	Claims Reported (b)	IBNR claim count (a) – (b)
2018	1,000	1,000	-
2019	1,100	1,100	-
2020	998	950	48
2021	2,100	1,600	500
2022	2,363	1,000	1,363
2023	1,181	250	931

Step2: Calculate the ultimate claim amount by multiplying Ultimate claim count with the average claim cost given.

Accident Year	Ultimate claim count (a)	Ultimate claim amount (b)= (a) X 200	Reported claim amount (c)
2018	1,000	2,00,000	1,60,000
2019	1,100	2,20,000	1,76,000
2020	998	1,99,500	1,52,000
2021	2,100	4,20,000	2,88,000
2022	2,363	4,72,500	1,90,000
2023	1,181	2,36,250	50,000

Subtract the reported claim amount given from the Ultimate claim amount to derive the IBNR claim amount.

Using the IBNR claim count and the average cost per claim given, pure IBNR claim amount is calculated.

IBNER is derived by subtracting pure IBNR claim amount from the IBNR claim amount.

Accident Year	IBNR claim amount (d)=(b)-(c)	IBNR Count (e)	Pure IBNR (f)=(e)x200	IBNER (d)-(f)
2018	40,000	-	-	40,000
2019	44,000	-	-	44,000
2020	47,500	48	9,500	38,000
2021	1,32,000	500	1,00,000	32,000
2022	2,82,500	1,363	2,72,500	10,000
2023	1,86,250	931	1,86,250	-

(Max 8)

iv) Ways to improve the estimation process.

- The split of IBNR is based on both claim count and claim amount projections.
- It is important to ensure that both the projection methods used are appropriate and consistent.
- The average cost per claim has been derived using the output of two independent projections. It is useful to compare and validate against other sources of information such as:
 - a relevant industry benchmark appropriate for home insurance
 - a long-term observed average for home insurance
- Appropriate adjustments to the IBNR estimation process would be required to account for:
 - claims inflation
 - any changes in the mix of business, policy terms and conditions, underwriting standards and claims handling procedures
 - any reason average cost of pure IBNR claims would be significantly different to other claims.
 - For example, pure IBNR claims could be significantly larger compared to IBNER and settled claims due to additional investigation and rigorous claims handling process.

(Max 2)

v) Bornhuetter-Ferguson (BF) method

Accident Year	Earned Premium (a)	Ultimate Claim Amount (b)	Chain Ladder Ultimate Loss Ratio (c)=(b)/(a) [Step 1]
2018	2,50,000	2,00,000	80%
2019	2,70,000	2,20,000	81%
2020	2,50,000	1,99,500	80%
2021	4,00,000	4,20,000	105%

2022	5,00,000	4,72,500	95%
2023	2,50,000	2,36,250	95%

Accident Year	Reported Claim Amount (d)	Cumulative Development Factor (e)=(d)/(b) [Step 2]	BF Ultimate Loss Ratio (c)x(e)+85%x(1-e) [Step 3]
2018	1,60,000	80%	81.00%
2019	1,76,000	80%	82.19%
2020	1,52,000	76%	81.04%
2021	2,88,000	69%	98.71%
2022	1,90,000	40%	88.82%
2023	50,000	21%	87.01%

Alternate steps:

*Step1: Calculate ELR ultimates = Prem*ELR*

Step 2: Calculate CDF (Developed%) using Ultimate / Reported

Step 3a: Use BF formula to calculate BF ultimates

Step 3b: Divide by premium to calculate BF LR

(Max 6)

vi)

- By nature, the Bornhuetter-Ferguson based ultimate loss ratio is a weighted average of the ultimate loss ratio from the chain ladder estimate and the expected ultimate loss ratio. The results of Bornhuetter-Ferguson can vary significantly from Chain ladder if the ultimate loss ratio assumption is significantly different to the chain ladder results:
 - The chain ladder estimate for 2020 accident year is closer to the expected loss ratio of 85% compared to other accident years.
 - For the later years, given the low development, significant weightage is given to the expected loss ratio. As a result, the ultimate loss ratio from Bornhuetter Ferguson is significantly lower compared to the chain ladder.
- Communication to the management should consider the key objectives of the exercise, technical knowledge of the audience...
 - ... ensuring clear representation of results with visual representation as suitable for effective communication.
- It is important to highlight the uncertainty in the estimation and key sources. Additional analysis might be required to quantify the uncertainty.
 - With the exception of 2020, it is observed that the chain ladder ultimate loss ratio is much higher than the expected loss ratio of 85%.
 - Compare results with older, fully developed cohorts.
 - Highlight actual vs expected differences from the previous reported figures to sensitise on the extent of deviation of results.
- A number of key assumptions were made for the calculation. It is important to highlight the critical assumptions, supported with logical and numerical evidence.
 - Expected loss ratio assumption chosen based on underwriting. Support with evidence to ensure suitability, adjusted for factors such as underwriting cycle, inflation etc.
 - Method of IBNR estimation used, compare with other methods like average cost per claim method
- Comment on the key limitations of the analysis
 - The rationale for choosing the same expected loss ratio for all accident years should be explained...

- ... supported with some sensitivity checks on using expected loss ratio from different sources.
- Accident year 2021 onwards, the IBNR is dominated by pure IBNR. Here there is heavy reliance on the average claim cost assumption of ₹200,000...
- ...This should be evidenced by results from alternative methods to check for impact of inflation, changes in business mix etc.
- Range of possible outcomes using stochastic methods and highlight rationale for best estimate used.

(Max 6)

[Max 28 Marks]

Q. 5) Adjustments to the reserving methodology

- i) Stop loss reinsurance can lead to lower net loss ratio compared to before if it exceeds the stop loss limit.
 - a. Use triangles gross of reinsurance for IBNR projection.
 - b. Calculate cohort level reinsurance recoveries later to derive net IBNR

(Max 1.5)

- ii) Waiting period clause would have resulted in lower claims in the initial development periods of earlier accident years. Some adjustments to consider:

- a. Exclude these periods for calculation of development factors as the waiting period doesn't apply to later accident years; or
- b. Modify the claim count of these cohorts to reflect a higher claims frequency, on par with periods where there is no waiting period

(Max 1.5)

- iii) Removal of discovery period can lead to increase in reporting delays for active accident years compared to those that have run-off.

- a. Consider a longer tail to the development factor to allow claim reporting beyond the current discovery period for an assumed proportion of claims (or)
- b. Increase in claim count to allow for those that might have been rejected earlier due to late reporting.

(Max 1.5)

- iv) Increase in claims handling speed leading to faster claim investigation and settlement, reducing claim handling expense.

- a. Separate projections for indemnity and expense portions of claims. Allow for reduced expense severity compared to prior accident cohorts
- b. Modify development factors to allow for quicker development in initial period and shorter tail.
- c. Since the system is new, there might be errors in claims settlement, and this may need to be allowed for.

(Max 1.5)

[Max 6 Marks]

Q. 6)

- i) Different classes of motor insurance (property/liability) are affected by inflation in different ways:
 - Motor property damage insurance responds mostly to the cost of parts,
 - Claims will tend to increase in line with general inflation.
 - A large proportion of motor claims cost is for the repair of vehicles...
 - ...will be affected by the level of earnings since these determine labour costs.

- Motor third party liability is often subject to higher levels of inflation, as there is a trend to more generous compensation in many markets.
- However, this may arise in steps rather than as a continuous process of inflation, as landmark legal judgements are handed down or legal reform comes into effect.
- Price inflation (body parts)
- Wage inflation (Bodily Injury, labour cost for repairs)
- Court awards / Legal inflation (including cost for lawyers, etc)
- Health / Medical inflation (Bodily injury)

(Max 2)

ii) Analyses that could be carried out to understand the financial impact of landmark legal judgement on company's claim reserves:

- Number of existing open and IBNR claims.
- Size of existing open and IBNR claims to understand the total size of claims with potential exposure to reserve change.
- Relevance of the landmark judgement to the company's open claims to understand the degree of exposure to specific case affected by landmark judgement.
- Degree of adoption of landmark judgement by other court cases.
- Proportion of claims settled via court vs out of court in the past.
- Existing difference in award between court settlement claims and out of court settled claims. Expect court settlements to go up if the difference in awards is now significant.
- Similarly, difference in settlement delay between court settled cases and out of court settled cases. Expect court settlements to go up as additional delays mean higher award.
- Extent of higher claims handling expenses as a result of additional legal fees.
- Past percentage of reopened claims during similar landmark judgements with adjustment for trends like increasing awareness.
- Analysis of emerging trends in liability claims:
 - i. "Jurisdiction shopping" resulting in claims being transferred to courts offering higher compensation.
 - ii. Increasing "No-win-no-fee" arrangements leading to more challenging and expensive litigation process.
 - iii. Higher fraud from staged accidents due to more attractive rewards.
- Extent of risk protection from current reinsurance arrangement and possibility of increase in future reinsurance premiums.
- Sense check against any guidance from country's actuarial body and regulator.
- Speed of settlement of existing open claims and impact of interest.
- Difference in assumed average claim cost vs average claim cost from judgement.
- Compare analyses with other landmark judgements from the past. There could be differences depending on extent of difference in landmark judgement vs current practice

(Max 7)

iii) Steps the company can take:

- Proactive measures to optimize claims process and reduce settlement delays.
- Consider increasing headcount for claims assessors.
- Make efforts to increase out of court settlements...
- ...compromise on existing cases under litigation, where seen beneficial.
- Increase reinsurance support – surplus and individual excess of loss to protect new business and renewals.
- Consider portfolio excess of loss and stop loss to protect overall book
- Reduce motor insurance capacity and divert capital to other lines of business until there is more stability
- Increase premiums to reflect additional claims costs
- Consider loss portfolio transfer of run off business if the expected impact poses solvency risk

- Purchase an adverse development cover if the cost is cheaper than the company's assumption.
- Invest reserves in assets that generate higher return, although this can be riskier.

(Max 5)

[Max 14 Marks]
